

MultiGridBarrier.jl: quasi-optimal solvers for convex variational problems

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

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Editor: 

Submitted: 09 June 2026

Published: unpublished

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Summary

MultiGridBarrier.jl is a Julia ([Bezanson et al., 2017](#)) package for solving convex variational problems in function spaces. These are the nonlinear partial differential equations (PDEs) and boundary-value problems that arise from minimizing a convex functional. Representative examples include the p -Laplacian for any $p \in [1, \infty]$, total-variation problems, and obstacle problems. The most useful of these are *nonsmooth*: the energy is convex but not differentiable (for example $p = 1$, or total variation), a regime in which Newton-type solvers applied naively either fail or require a number of iterations that grows rapidly with the mesh resolution.

The package implements the **multigrid barrier method** ([Loisel, 2020, 2026b, 2026e](#)), which couples an interior-point (barrier) method with a multigrid hierarchy. For the problem classes covered by the supporting theory the method is *quasi-optimal*: the number of interior-point/Newton iterations grows only mildly with the number of degrees of freedom n . For instance, this count is $O(\sqrt{n} \log n)$ for the p -Laplacian ([Loisel, 2020](#)), and polylogarithmic in the analytic, spectral setting ([Loisel, 2026e](#)).

MultiGridBarrier.jl provides finite-element discretizations in one, two, and three dimensions (simplicial P_1/P_2 elements and tensor-product Q_k elements), as well as Chebyshev spectral discretizations, all with isoparametric element maps. It builds an algebraic-multigrid hierarchy automatically, supports user-specified mesh connectivity (enabling slit domains, branch cuts, and glued manifolds), solves time-dependent problems, and offers optional GPU acceleration through CUDA. A typical solve is three lines:

```
using MultiGridBarrier
geom = fem2d_P2()
sol = mgb_solve(assemble(amg(geom); p = 1.0)) # a nonsmooth p = 1 problem
```

Statement of need

Convex variational problems are ubiquitous in computational science: nonlinear elasticity and plasticity, image denoising and segmentation (total variation), contact and obstacle problems, and non-Newtonian flow (the p -Laplacian). The difficulty is that the most interesting cases are nonsmooth (the energy is convex but not differentiable), so Newton-type methods applied naively either stagnate or require an iteration count that grows rapidly as the mesh is refined.

Interior-point (barrier) methods handle nonsmoothness robustly by following a smooth central path, but a single barrier solve still requires solving a sequence of large, increasingly ill-conditioned linear systems. The multigrid barrier method addresses both issues at once: a multigrid hierarchy preconditions the central-path subproblems so that the *total* cost stays close to linear in the number of unknowns, with rigorous bounds for the covered problem classes ([Loisel, 2020, 2026b, 2026e](#)).

37 State of the field

38 General-purpose finite-element libraries in Julia, such as Gridap.jl (Badia & Verdugo, 2020)
 39 and Ferrite.jl (Carlsson et al., 2024), and in the wider ecosystem, such as the FEniCS project
 40 (Alnæs et al., 2015), provide flexible tools for discretizing PDEs; algebraic-multigrid libraries
 41 such as AlgebraicMultigrid.jl (JuliaLinearAlgebra contributors, 2024) and PyAMG (Bell et al.,
 42 2023) solve the resulting linear systems. These are general building blocks, but none provides
 43 an out-of-the-box, theoretically grounded solver for *nonsmooth convex variational* problems.
 44 MultiGridBarrier.jl fills this gap. It packages a discretization, a multigrid hierarchy, and
 45 a barrier solver behind a small high-level interface, and builds on (rather than reinvents) the
 46 Julia ecosystem: it uses AlgebraicMultigrid.jl, or optionally PyAMG, to coarsen its auxiliary
 47 problems, and runs on both CPU and GPU.

48 Software design

49 A problem is solved with four composable steps. A *mesh constructor* (fem1d, fem2d, fem2d_P1,
 50 fem2d_P2, fem3d, spectral1d, spectral2d) returns a Geometry describing the discretization.
 51 amg(geom) attaches an algebraic-multigrid hierarchy, assemble(mg; ...) builds the convex
 52 problem (the functional, its barrier, and any constraints), and mgb_solve(prob) runs the
 53 barrier method.

54 Internally, the multigrid barrier method tracks the central path of an interior-point method
 55 while using a multigrid hierarchy to solve the Newton systems along that path (Loisel, 2026b,
 56 2026e). The finite-element discretizations are isoparametric simplicial P_k and tensor-product
 57 Q_k elements. For these, the multigrid hierarchy is built algebraically: the package coarsens an
 58 auxiliary P_1/Q_1 problem on the element corners and lifts the resulting transfer operators to the
 59 full high-order basis, so the same machinery serves every finite-element family. Mesh topology
 60 is represented by an explicit node connectivity array, which decouples geometry from topology
 61 and lets geometrically coincident nodes remain distinct, supporting slit domains, branch cuts,
 62 and glued manifolds. The package also provides a structured, batched-GEMM assembly of the
 63 Newton Hessians that maps efficiently onto GPUs (Loisel, 2026a), a variant for nonuniform
 64 grids (Loisel, 2026d), and the algebraic formulation used here (Loisel, 2026c).

65 The Chebyshev spectral discretizations (spectral1d, spectral2d) use an intrinsically spectral
 66 hierarchy instead. The multigrid levels are a sequence of polynomial approximation spaces of
 67 increasing degree, with exact polynomial interpolation as the inter-level transfer and zero-trace
 68 boundary conditions imposed by basis construction rather than node masking. This is the
 69 setting of the spectral barrier method (Loisel, 2026e): when the solution is analytic, the
 70 spectral discretization converges geometrically and the overall method is quasi-optimal, with
 71 the iteration count growing only polylogarithmically in the number of degrees of freedom.

72 Convex constraints (p-norm / Euclidian-power, linear, and piecewise) are built in and composable
 73 for any discretization, and parabolic_solve extends the method to time-dependent problems.

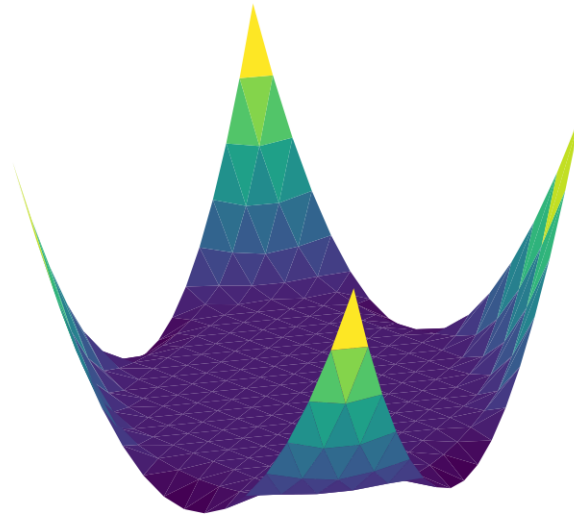


Figure 1: A nonsmooth ($p = 1$) solution of a two-dimensional p -Laplace problem, computed with MultiGridBarrier.jl on a refined Q_2 mesh and rendered with the package's plotting front-end.

Research impact

The methods implemented in MultiGridBarrier.jl underpin the numerical experiments in the papers that introduce them (Loisel, 2020, 2026b, 2026e), and the solver and its earlier implementations have been used by other researchers. For example, Zhang and Jiang use this algorithm within a convolutional-neural-network reduced-order modeling method for multiscale problems (Zhang & Jiang, 2025). More broadly, the underlying p -Laplacian algorithm (Loisel, 2020) has been cited around 26 times (Google Scholar, as of June 2026) across the numerical-PDE and optimization literature, in areas such as computational p -Laplacian numerics (Balci et al., 2023) and p -harmonic shape optimization (Müller et al., 2021).

AI usage disclosure

Generative AI assistance (Anthropic Claude, via the Claude Code command-line tool) was used in preparing this submission: drafting and editing this paper and parts of the documentation and README, generating the illustrative figure, and assisting with some software changes (refactoring, a correctness fix, and a mesh-connectivity feature). All AI-assisted output was reviewed, validated, and edited by the author, who made all core design and research decisions and takes full responsibility for the software and the manuscript.

Acknowledgements

This work received no specific external funding. The author declares no conflicts of interest.

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